Ethan Crawford

GGIS 478 Final Report

May 12th, 2026

Script link

Multi-sensor remote sensing of restoration trajectories at a small Midwestern land trust

Sentinel-2, Sentinel-1 SAR, and Random Forest classification at Warbler Ridge Conservation Area, 2017-2024

Abstract

Small land trusts protect a lot of restored land across the Midwest, but they rarely have the resources to monitor restoration outcomes systematically. This project asks whether free satellite imagery and a simple machine learning workflow can fill that gap. I used Sentinel-2 optical imagery and Sentinel-1 synthetic aperture radar (SAR) to map land cover at Warbler Ridge Conservation Area (WRCA), a 447-hectare property in Coles County, Illinois owned by Grand Prairie Friends, for five years between 2017 and 2024. For each year I built a 19-band feature stack (10 Sentinel-2 reflectance bands, 6 spectral indices, and 3 winter SAR bands) and trained a Random Forest classifier with four classes: water, forest, cropland, and prairie. The classifier reached 94.1% overall accuracy with a Kappa of 0.91 when evaluated on held-out polygons. Sentinel-1 VH backscatter was the single most important feature in the classifier, suggesting that radar adds information that optical bands alone don't capture. Comparing 2017 and 2024 within the property, I found 40.5 ha of cropland-to-prairie change and 47.0 ha of forest-to-other change, both consistent with documented prairie plantings and timber stand improvement work. Only 0.6 ha of cropland-to-forest change was detected, which probably reflects the fact that trees planted in 2020 had not yet grown a closed canopy by 2024. The

workflow shows that small land trusts could monitor their own restoration work using only free data and tools.

Introduction

Less than 0.01% of Illinois' original tallgrass prairie remains today, and a lot of the river corridors, wetlands, and forests across the Midwest have been altered by agriculture. Land trusts do much of the on-the-ground restoration work on the landscapes that are left, but the time and money they spend on acquisition and management usually leaves little room for systematic monitoring. Free satellite data from the Sentinel-1 and Sentinel-2 missions, combined with cloud platforms like Google Earth Engine, offers a way to fill that gap (Gorelick et al., 2017).

Random Forest has become one of the most common ways to classify land cover from satellite imagery. It works well with many input features at once, gives interpretable variable importance scores, and tends to avoid overfitting (Breiman, 2001; Belgiu & Drăguţ, 2016). Combining optical and SAR imagery is also widely used because the two sensors measure different things: optical captures reflected sunlight, while SAR measures how microwaves bounce off the land surface. Pairing them often improves accuracy in mixed agricultural-natural landscapes (Clerici et al., 2017; Moharrami et al., 2024).

This project asks whether this kind of multi-sensor satellite workflow can detect and characterize a decade of restoration at Warbler Ridge Conservation Area (WRCA), a property managed by Grand Prairie Friends (GPF) in east-central Illinois. The three goals are: (1) to see whether SAR features actually contribute information beyond what optical bands provide; (2) to map cover class changes between a 2017 baseline and 2024; and (3) to compare what the classifier detects against the restoration activities GPF has published in its annual reports.

Study Area

Warbler Ridge Conservation Area covers about 447 ha in Coles County, Illinois, roughly 3.5 miles south of Charleston. The property runs along the Embarras River and is directly next to Fox Ridge State Park, so it forms part of a longer protected river corridor. Five sub-units inside WRCA are protected as Illinois Nature Preserves or Land and Water Reserves. GPF assembled the property piece by piece between 2012 and 2024 — starting with the 141-acre Embarras Ridges parcel in 2012, then adding the 193-acre Warbler Woods Nature Preserve (an 1850s-era remnant forest) in 2014, a 274-acre acquisition in 2018, and Warbler Bottoms North (92 acres) in 2019.

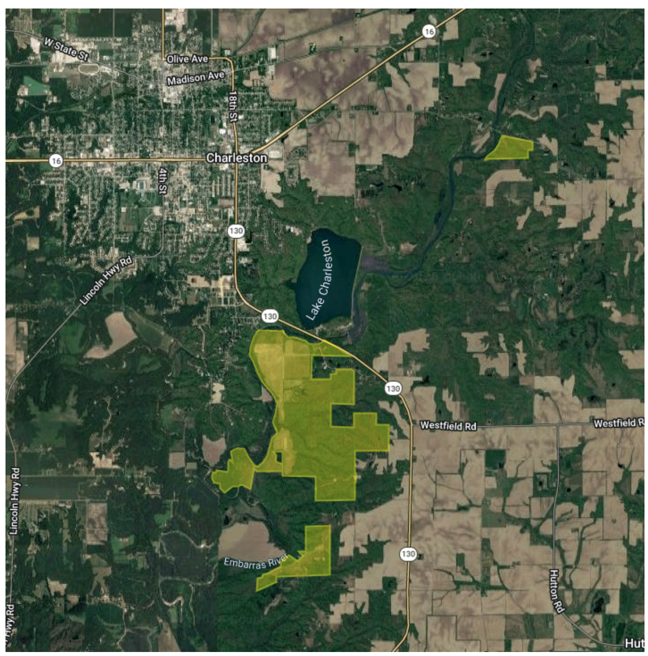

Figure 1: Study area map. Warbler Ridge Conservation Area boundary on NAIP 2023 backdrop, with state and county inset showing location in east-central Illinois.

The land cover inside WRCA is a mix of floodplain and upland forest, restored wetlands, established and developing prairie/pollinator plantings, and a small amount of cropland that is being transitioned out. The surrounding matrix is almost entirely row-crop agriculture in a cornsoybean rotation, which is why training polygons for the cropland class had to be drawn outside the property boundary.

Restoration activity at WRCA during the study window was substantial. According to GPF's published management timeline, work between 2015 and 2024 included 141 acres of maple thinning and invasive removal at Embarras Ridge (2015); 20 acres of pollinator planting (2016); 99 acres of combined invasive removal, timber stand improvement (TSI), and reforestation (2017); 9 acres of wetlands construction (2018); 10 acres of bat habitat restoration with 2,000+ trees plus 60 acres of cover crop and a 3-acre wetland (2019); 50,000 trees planted on 40 acres protected by deer fence, with 42 more acres of cover crop and 95 acres of TSI (2020); and additional TSI and brush removal in 2024 covering roughly 192 acres in total (100 acres of upland TSI, 17 acres of river bottom thinning, and 75 acres of invasive brush removal). The 2020 reforestation campaign was part of GPF's bat habitat restoration project, which targets habitat for the federally endangered Indiana bat and the threatened Northern Long-Eared bat.

Data and Methods

Imagery and Reference Data

I used Sentinel-2 surface reflectance imagery (the COPERNICUS/S2_SR_HARMONIZED collection in Google Earth Engine) as the optical part of the feature stack. For each of the five study years (2017, 2019, 2021, 2023, 2024), I pulled all scenes intersecting the property boundary between June 1 and August 31, which is peak growing season in this part of Illinois. To handle clouds, I joined each scene to the s2cloudless cloud probability product by system index and masked any pixels with cloud probability greater than 30%. For the radar part of the stack, I used Sentinel-1 Ground Range Detected (GRD) imagery

(COPERNICUS/S1_GRD) in interferometric wide (IW) mode with VV and VH dual polarization. I pulled SAR scenes for December through February of each focal year, restricting to the ascending orbit pass because the descending orbit had no usable coverage of the study area. NAIP imagery at 1-meter resolution was used for drawing training polygons by visual inspection, and GPF's annual reports provided the acquisition and management timeline.

Boundary and Training Region

The property boundary was digitized in QGIS from GPF's published parcel maps and uploaded to Earth Engine as a Feature Collection. The four constituent features were dissolved into a single 446.8 ha polygon. Because the cropland class doesn't really exist inside WRCA (it's mostly forest and prairie inside the boundary) cropland training polygons had to be drawn in the surrounding agricultural matrix. To make sure these outside-the-boundary points still got pixel values during training, I gave the 2024 training feature stack a 5 km outward buffer of the original AOI. However, the classification feature stacks for the five study years were still built only over the unbuffered AOI.

Optical Features

For each year and region, the cloud-masked Sentinel-2 scenes within the June-August window were combined into a per-pixel median composite using ten reflectance bands (B2 through B12, excluding the cirrus and water-vapor bands). From the median composite I then computed six spectral indices. The choice of indices reflects what these bands actually measure on the ground. Healthy green vegetation absorbs red and blue light strongly through chlorophyll in the palisade layer and scatters near-infrared (NIR) light through the spongy mesophyll, so any ratio involving red or blue and NIR is sensitive to vegetation vigor (Gitelson & Merzlyak, 1996). The six indices are: NDVI for general greenness, EVI for high-biomass canopies where NDVI saturates, NDWI (using green and NIR) for water bodies, NDMI for leaf moisture using the SWIR-NIR contrast (Hunt, 1989), NDRE for red-edge chlorophyll dynamics, and NBR (NIR-SWIR contrast) which helps separate dense vegetation from bare or disturbed surfaces because of its ability to detect woody structure differences. The final optical stack for each year contained 16 bands.

SAR Features

The Sentinel-1 scenes for the December-February winter window were first run through a speckle filter at a 30 m radius, then combined into a per-pixel median. From the winter median I computed three SAR features: VV backscatter (dB), VH backscatter in (dB), and the VV/VH ratio (calculated as VV minus VH because both are already log-scaled). Using winter SAR is a deliberate choice. In the dormant season, deciduous forest is leaf-off but tree trunks and large branches still scatter microwaves and produce high VH backscatter. Cropland is bare or stubble-covered in winter, so it produces low backscatter in both polarizations. Open water reflects microwaves specularly away from the sensor and shows up as very low backscatter. These differences let SAR cleanly separate the consistent woody vegetation from temporary cover types during a season when the optical imagery is often going to be blocked by clouds anyway.

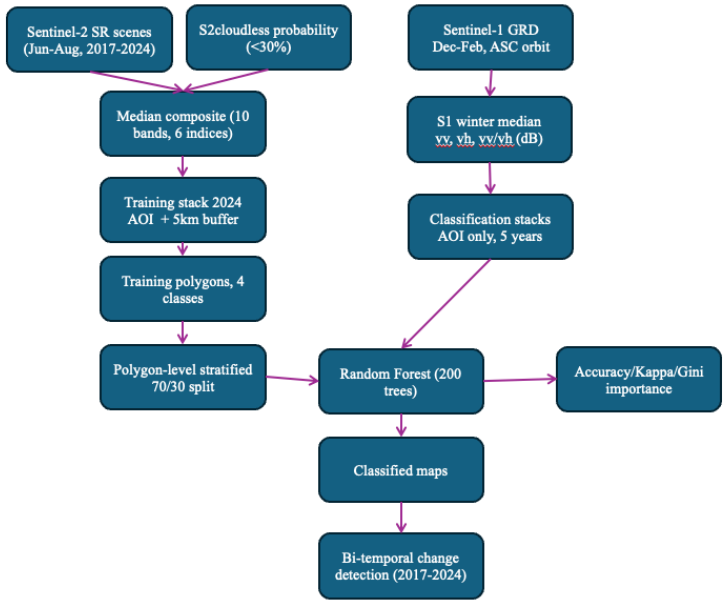

Figure 2: Methodology flowchart

Training Data and Random Forest

I drew training polygons by hand, primarily using NAIP and SAR layers as a reference, producing 66 polygons across four cover classes: water, forest, cropland, and prairie. A fifth class for bare ground was originally included but had to be dropped because the total polygon area was too small to sample from, and bare ground was sparse at the site anyway. Then, within each polygon, random points were generated, distributed roughly evenly across the polygons in each class so that smaller polygons still contributed. Each point was tagged with both its class code and the ID of the polygon it came from. This follows the training data best-practices, at least 100 pixels per class and homogeneous training sites.

I sampled the full 19-band 2024 training stack at each training point location at 10 m scale. After trial-and-error producing incomplete results, I decided to split into training and test sets. I used a polygon-level stratified 70/30 split. This is more conservative than the typical pixel-level split: because pixels within a single polygon are spatially close and tend to look very similar to each other, splitting at the pixel level could let the classifier "memorize" polygons and inflate accuracy. So by splitting at the polygon level, it guarantees that the test set contains only pixels from polygons the classifier has never seen. The stratification by class made sure all four classes appeared in both train and test sets.

I then trained a Random Forest classifier with 200 trees using all 19 features as inputs. Random Forest is a collection of many decision trees. Each tree is similar to the CART classifier, but the algorithm trains many of them on bootstrap samples and uses random subsets of features at each split, then averages the predictions. This level of averaging is what makes RF less prone to overfitting than a single tree, and it's also what produces the Gini importance scores that I used to evaluate feature contributions and evaluate my methodology. The trained classifier was then applied to each of the five annual feature stacks within the AOI.

Change Detection

Change detection here was a simple post-classification approach. For each pixel inside the AOI, I compared the 2017 class label to the 2024 class label and recorded any difference as a transition. Three transitions were of particular interest given the documented management activity at WRCA:

- cropland to forest (the expected result of completed reforestation)
- cropland to prairie (the expected result of prairie or pollinator planting)
- forest to other (the expected result of canopy thinning under TSI or brush removal).

For each transition, I computed the total area within the AOI. Validation was done in two ways: internally, using the held-out 30% test polygons to produce a confusion matrix and overall accuracy and Kappa; and externally, by comparing the detected transition areas against the activities GPF has published in its annual reports and management timeline.

Results

Classification Accuracy

The Random Forest classifier achieved an overall accuracy of 94.1% and a Kappa coefficient of 0.913 on the polygon-level test set. The test set contained 102 points across 16 polygons, with all four classes represented. The confusion matrix (Table 1) showed that water, forest, and prairie classes were classified without errors on the test set. The only confusion was in the cropland class, where six pixels (12.5% of cropland test points) were classified as prairie. This kind of confusion makes sense. Some crops, especially soybean fields in late summer, could look spectrally similar to warm-season prairie grasses.

	Water	Forest	Cropland	Prairie
Water (truth)	24	0	0	0
Forest (truth)	0	6	0	0
Cropland (truth)	0	0	42	6
Prairie (truth)	0	0	0	24

Table 1. Confusion matrix for the 4-class Random Forest classifier on the polygon-level held-out test set. Rows are reference (truth); columns are predicted.

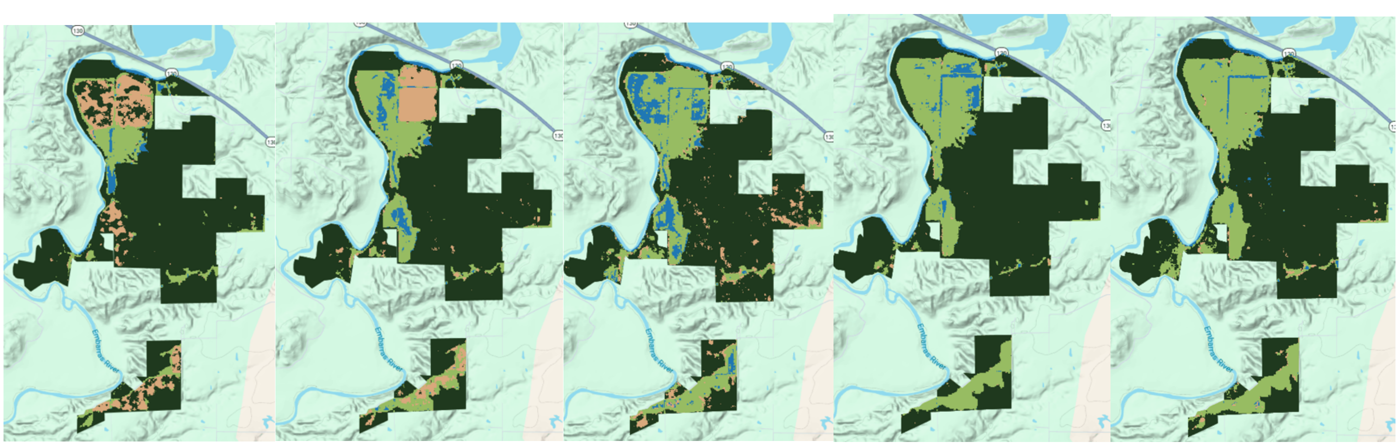

Figure 3: Year-by-year classification maps for 2017, 2019, 2021, 2023, and 2024 (left to right). Color scheme: water (blue), forest (dark green), cropland (tan), prairie (light green).

Feature Importance

The Gini importance scores from the trained classifier put Sentinel-1 VH backscatter as the single most important feature, followed by NDRE, NDVI, NDWI, and the Sentinel-2 blue band (Table 2). Two of the top six features are SAR-derived (S1_VH at rank 1 and S1_VV_VH at rank 6). This is a useful finding because it directly supports adding SAR to the classifier. The radar bands are not redundant with the optical bands. They're contributing something the optical bands can't, which is structural information about woody vegetation versus open ground.

Rank	Feature	Gini Importance
1	S1_VH	38.86
2	NDRE	34.46
3	NDVI	33.16
4	NDWI	30.72
5	B2 (blue)	27.65
6	S1_VV_VH	24.66
7	B5 (red edge 1)	19.17
8	EVI	19.06
9	B6 (red edge 2)	18.35
10	NBR	18.02

11		

Table 2. Top ten features by Gini importance from the trained Random Forest classifier.

Change Detection

Applying the trained classifier to all five annual stacks and comparing 2017 with 2024 within the AOI gave the three transition areas in Table 3. Cropland to prairie came in at 40.5 ha, which is in the same range as the combined 102 acres (41 ha) of cover crop work GPF reported in 2019 and 2020 plus the 20 acres of pollinator planting from 2016. Forest to other was 47.0 ha, broadly consistent with the 95 acres (38 ha) of TSI documented in 2020 plus the 17 acres of river bottom thinning in 2024. Some fraction of the forest-to-other signal is probably classifier noise between years rather than real change, which is normal for bi-temporal classification.

Transition	Area (ha)	GPF management activity
Cropland to Forest	0.6	40 ac reforestation in 2020
Cropland to Prairie	40.5	102 ac cover crop (2019-2020) + 20 ac pollinator prairie (2016)
Forest to Other	47	95 ac TSI in 2020 + 17 ac thinning in 2024

Table 3. Bi-temporal change detection within the AOI (2017-2024).

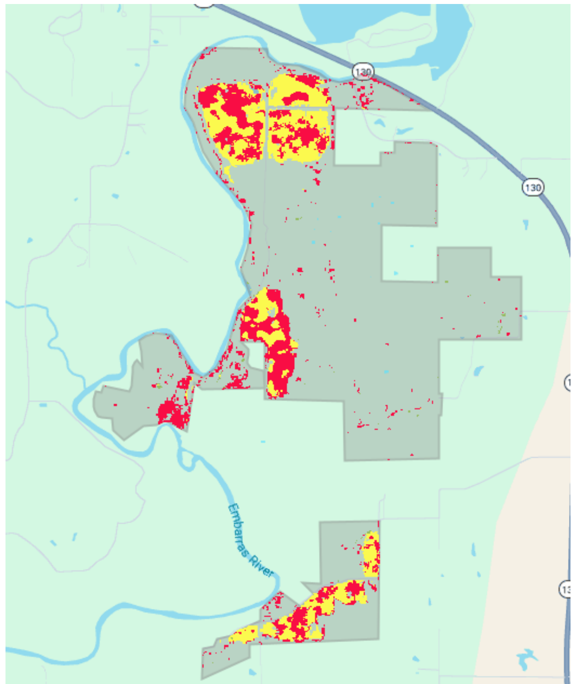

Figure 4: Transition map for 2017-2024 within the AOI, with cropland-to-prairie (yellow), cropland-to-forest (green), and forest-to-other (red)

The most striking result is the very low cropland-to-forest area (0.6 ha) compared to what the 2020 reforestation campaign would suggest if all the trees had matured into closed canopy by 2024. Two potential reasons for this gap are discussed below.

<u>Discussion</u>

SAR Contributes Real Information

The classifier reached 94.1% accuracy with a polygon-level split. This is a stricter test than a pixel-level split, which is the more common default but tends to inflate the accuracy when adjacent pixels from the same polygon end up in both train and test (autocorrelation). A Kappa of 0.91 falls comfortably in the 'substantial agreement' range used in the remote sensing literature (Belgiu & Drăguţ, 2016). The more interesting result is the feature importance ranking: Sentinel-

1 VH backscatter was the single most important feature, and another SAR feature (the VV/VH ratio) was in the top six. This is consistent with what Clerici et al. (2017) and Moharrami et al. (2024) found in other landscapes. SAR contributes information about vegetation structure that optical bands don't capture, especially in winter when leaves are off and the contrast between standing trees and bare cropland is sharpest.

The Cropland-to-Forest Discrepancy

The 47.0 ha of forest-to-other change is in the same order of magnitude as the documented 95 acres (38 ha) of TSI from 2020 plus the 17 acres of river bottom thinning in 2024. The 40.5 ha of cropland-to-prairie change matches up well with the roughly 102 acres of cover crop work GPF did in 2019 and 2020 (plus some additional pollinator planting from 2016). The classifier seems to be picking up the right ecological signals for those two transitions.

The cropland-to-forest signal at 0.6 ha is much lower than the 2020 reforestation campaign (50,000 trees on 40 acres) would predict if all the plantings had matured into recognizable forest by 2024. There are two plausible reasons, and they probably both contribute.

The first is ecological. Trees planted in 2020 were only four years old in 2024. At 10 m spatial resolution and in mid-summer Sentinel-2 imagery, a pixel over a young plantation captures a mixture of widely-spaced young saplings and the cool-season or warm-season grass cover between them. The dominant spectral signal at that age is grass, not canopy. This is actually a classic example of what the vegetation remote sensing lecture flagged as one of the biggest challenges in the field: a young plantation is a leaf-scale forest sitting inside a canopy-scale pixel. The pixel correctly reads as prairie even though management intent was reforestation. The literature on post-planting recovery using satellite data finds that canopy closure indicators typically lag actual planting by 8 to 15 years (White et al., 2018), so a four-year-old plantation reading as grass is exactly what would be expected. Under this reading, the cropland-to-prairie transition area absorbs most of what was nominally planted as forest.

The second reason, and honestly the most likely, is methodological. My forest training polygons were small and were drawn over the interiors of mature canopy stands. The classifier may have learned a narrow definition of 'forest' that corresponds to closed-canopy mature deciduous forest specifically, rather than younger woody regrowth. If I'd included examples of young plantings as a dedicated training class, more of the cropland-to-forest signal might have been recovered. Some of the cropland-to-prairie classification could also be incorrectly classified, and in reality is cropland-to-forest. However, since the training data didn't include young plantings as their own class, this study can't fully separate the ecological from the methodological explanation.

Limitations

A few things limited what this study can conclude. The Sentinel-2 archive only starts in March 2017, so the 2017 baseline is contemporaneous with GPF's earliest acquisitions rather than predating them. By 2017 there had already been substantial restoration activity on the property (141 ac of maple thinning in 2015, 20 ac of pollinator planting in 2016, and 99 ac of mixed work in 2017 itself), so some of what 'changed' between 2017 and 2024 was already in progress at the baseline. The 10 m pixel size produces some level of edge-mixing at parcel boundaries and in narrow features like riparian forest strips. The four-class scheme is necessarily coarse for the site. Separating young woody from established prairie would have been useful but would have required training polygons that didn't exist when I started. Training polygons were drawn from imagery rather than ground-truthed, so any visual interpretation bias then propagates into the classifier. Finally, the lack of per-parcel validation against GPF's acquisition timeline limits the strength of the documented-event comparison.

Implications for Small Land Trusts

Small land trusts protect a meaningful fraction of restored land in the Midwest but don't usually have the staff or budget for systematic ecological monitoring. The workflow used here relies only on free satellite data (Sentinel-1 and Sentinel-2), runs entirely on the free Google Earth Engine platform, and validates against publicly available annual reports. With 50 to 100 manually digitized training polygons and a few hours of processing, the result is a decent classification with interpretable feature importance and a quantified before and after comparison. Significant remote sensing knowledge is still needed for core methodological decisions, but the basic pipeline could serve as a template for small organizations that want to demonstrate the impact of their conservation work to members, donors, or grant agencies.

Conclusion

Combining Sentinel-2 optical imagery with Sentinel-1 SAR and a Random Forest classifier can detect and characterize years of restoration at a small Midwestern conservation property. The 94.1% accuracy with a polygon-level evaluation, the dominance of SAR features in the importance ranking, and the close matches between detected transitions and documented management activities together suggest the workflow is doing the right thing. The unexpectedly low cropland-to-forest signal is itself may be useful finding; even four years after planting, young woody restoration on former cropland still reads as grass rather than canopy at this resolution, which matters for how reforestation is monitored from satellites. The pipeline shown here is replicable using only free data and publicly available reports, which suggests an accessible monitoring template for small land trusts.

References

Belgiu, M., & Drăguţ, L. (2016). Random forest in remote sensing: A review of applications and future directions. ISPRS Journal of Photogrammetry and Remote Sensing, 114, 24–31. https://doi.org/10.1016/j.isprsjprs.2016.01.011

Breiman, L. (2001). Random forests. Machine Learning, 45(1), 5–32. https://doi.org/10.1023/A:1010933404324

Clerici, N., Valbuena Calderón, C. A., & Posada, J. M. (2017). Fusion of Sentinel-1A and Sentinel-2A data for land cover mapping: A case study in the lower Magdalena region, Colombia. Journal of Maps, 13(2), 718–726. https://doi.org/10.1080/17445647.2017.1372316

Gitelson, A. A., & Merzlyak, M. N. (1996). Signature analysis of leaf reflectance spectra: Algorithm development for remote sensing of chlorophyll. Journal of Plant Physiology, 148(3–4), 494–500. https://doi.org/10.1016/S0176-1617(96)80284-7

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sensing of Environment, 202, 18–27. https://doi.org/10.1016/j.rse.2017.06.031

Grand Prairie Friends. (2019–2024). Annual reports and conservation updates. Grand Prairie Friends Land Trust.

Hunt, E. R., Jr. (1989). Detection of changes in leaf water content using near- and middle-infrared reflectances. Remote Sensing of Environment, 30(1), 43–54. https://doi.org/10.1016/0034-4257(89)90046-1

Moharrami, M., Attarchi, S., Gloaguen, R., & Alavipanah, S. K. (2024). Integration of Sentinel-1 and Sentinel-2 data for ground truth sample migration for multi-temporal land cover mapping. Remote Sensing, 16(9), 1566. https://doi.org/10.3390/rs16091566

White, J. C., Saarinen, N., Kankare, V., Wulder, M. A., Hermosilla, T., Coops, N. C., Pickell, P. D., Holopainen, M., Hyyppä, J., & Vastaranta, M. (2018). Confirmation of post-harvest spectral recovery from Landsat time series using measures of forest cover and height derived from airborne laser scanning data. Remote Sensing of Environment, 216, 262–275. https://doi.org/10.1016/j.rse.2018.07.004